HVAC & MEP INSTALLATION WORK

Chillers, AHUs, Electrical Panels & Heat Exchangers

This document provides a professional overview of installation, commissioning, operation, maintenance and practical uses of major HVAC and MEP equipment. It is suitable as a project portfolio or service capability document for HS Bukhari Engineering & HVAC Services.

1. Chiller Installation & Applications

A chiller is a central cooling machine used to produce chilled water for air-conditioning and process-cooling applications. Chillers are commonly installed in commercial buildings, hospitals, universities, hotels, factories, laboratories and other large facilities.

Installation Activities

- Site inspection, equipment positioning and foundation preparation.
- Installation and alignment of the chiller on a suitable base with proper vibration control.
- Connection of chilled-water supply and return piping with valves, strainers, flexible connectors and insulation.
- Connection of condenser-water piping for water-cooled chillers, where applicable.
- Electrical power connection, earthing and control wiring.
- Installation and checking of pumps, flow switches, temperature sensors and pressure instruments.
- System flushing, pressure testing, leak inspection and water treatment.
- Commissioning, set-point verification and performance testing.

Common Uses

- Central air-conditioning for large commercial and institutional buildings.
- Cooling of AHUs and FCUs through chilled-water circuits.
- Process cooling for industrial equipment.
- Temperature control in laboratories, healthcare and technical facilities.

Key Maintenance Points

Regular maintenance includes checking chilled-water temperatures, pressures, flow, strainers, pumps, refrigerant or condenser-water conditions, electrical connections, alarms, vibration and overall operating efficiency.

2. AHU INSTALLATION & HVAC AIR DISTRIBUTION

An Air Handling Unit (AHU) conditions and distributes air through a building. Depending on the system design, an AHU can provide filtration, cooling, heating, humidity control, fresh-air mixing and air distribution.

AHU Installation Activities

- Positioning the AHU on a suitable level base or support frame.
- Connecting chilled-water or DX cooling components according to the system design.
- Connecting supply, return and fresh-air ducts with proper flexible connections.
- Installing filters, dampers, actuators, sensors and control devices.
- Connecting drain piping and providing a correctly trapped condensate drain.
- Electrical wiring for fans, motors, controls and auxiliary equipment.
- Insulating ductwork and chilled-water piping where required.
- Balancing airflow and checking supply and return air temperatures.

AHU Uses

- Air conditioning and ventilation in hospitals, offices and educational buildings.
- Clean and filtered air distribution in laboratories and controlled areas.
- Fresh-air ventilation and indoor air-quality management.
- Temperature control for commercial and industrial spaces.

Commissioning & Service

During commissioning, technicians check fan rotation, motor current, airflow, filter condition, chilled-water flow, coil performance, drain operation, damper movement, sensor readings and control-system response. Proper commissioning helps maintain comfort, reliability and energy efficiency.

Typical AHU Maintenance

Routine work includes filter cleaning or replacement, coil cleaning, blower and motor inspection, bearing checks, belt tension, drain-tray cleaning, actuator testing, electrical inspection and temperature verification.

3. ELECTRICAL PANEL INSTALLATION & CONTROL

Electrical panels distribute electrical power and provide protection and control for HVAC and MEP equipment. They may contain breakers, contactors, overload relays, control relays, terminal blocks, meters, timers, PLC or BMS interfaces and other control components.

Panel Installation Activities

- Reviewing the electrical drawing, load schedule and panel location.
- Mounting the panel securely and maintaining required working clearance.
- Installing breakers, contactors, overload relays, control components and terminals.
- Routing and dressing incoming and outgoing cables.
- Ferruling, labeling and terminating cables correctly.
- Providing protective earthing and checking continuity.
- Testing insulation resistance and verifying circuit identification.
- Checking phase sequence, voltage, control logic and protective-device operation.
- Connecting HVAC equipment such as chillers, pumps, AHUs, fans and heat-exchange systems.

Applications

- Power and control of HVAC equipment.
- Motor starting and protection for pumps and fans.
- Automatic control of contactors, valves and other field devices.
- Integration with BMS or other building control systems.
- Electrical protection, isolation and monitoring.

Safety & Testing

Electrical panel work should be carried out by qualified personnel using appropriate isolation and safety procedures. Before energizing a panel, connections, earthing, protective devices, cable terminations and control circuits should be inspected and tested.

4. HEAT EXCHANGER INSTALLATION & SYSTEM USE

A heat exchanger transfers thermal energy between two fluids without normally mixing them. Heat exchangers are used in HVAC, hot-water systems, process cooling, heating systems and industrial applications.

Installation Activities

- Selecting a suitable location with adequate access for inspection and maintenance.
- Installing the unit on a suitable support and ensuring correct alignment.
- Connecting primary and secondary flow and return piping.
- Installing isolation valves, strainers, gauges, temperature sensors and control valves.
- Providing suitable drainage and air-venting arrangements.
- Flushing the piping system and checking for leaks.
- Checking flow direction and verifying operating pressures and temperatures.
- Insulating the exchanger and associated piping where required.
- Commissioning the system and confirming heat-transfer performance.

Common Uses

- Heating and cooling water transfer in HVAC systems.
- Domestic hot-water generation.
- Chilled-water and condenser-water applications.
- Industrial process heating and cooling.
- Energy recovery and heat-recovery systems.

Integrated HVAC & MEP Site Work

Professional site work often requires coordination between chillers, AHUs, pumps, electrical panels, heat exchangers, control systems and piping. Correct installation depends on accurate equipment positioning, proper pipe and cable routing, safe electrical connections, adequate insulation, commissioning and systematic maintenance.

Professional Service Scope

- Installation and replacement of HVAC and MEP equipment.
- Preventive and corrective maintenance.
- Electrical panel wiring and troubleshooting.
- AHU, chiller, pump and motor servicing.
- Heat-exchanger installation and maintenance.
- Testing, commissioning and technical inspection.

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